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Enhancing knowledge tracing with multi-level individualized perception and teacher-student semantic distillation

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ABSTRACT

Knowledge tracing aims to model students' dynamic knowledge states based on their historical learning interactions and predict future learning performance. Existing sequence modeling methods often overlook individual differences and have limited semantic modeling capability. To address these issues, this paper proposes a knowledge tracing model that combines personalized modeling with knowledge distillation. The model introduces three personalized modules: (1) a personalized question understanding module that captures individual differences in students' understanding of the same question; (2) a personalized question-knowledge association module that models relationships between questions and relevant knowledge concepts; and (3) a personalized knowledge state forgetting module that simulates students' memory decay patterns. These modules allow for more accurate modeling of students' dynamic knowledge states. Furthermore, to overcome the semantic limitations of lightweight models, a large language model (LLM) is used as the teacher, and its semantic modeling capability is transferred to an LSTM-based student model via knowledge distillation. Experiments show that the proposed method consistently improves prediction performance on two benchmark datasets, demonstrating its effectiveness in modeling personalized learning and enhancing semantic representation.

1. Introduction

With the development of computer-assisted education (CAE) systems, online learning platforms have accumulated large amounts of learner interaction data. These data provide an important foundation for personalized instruction and learning support (Conati et al., 2021; VanLehn et al., 2017). Knowledge Tracing (KT) aims to model learners' dynamic knowledge states from their historical interactions and predict their performance in future learning activities (Abdelrahman et al., 2023; Shen et al., 2024). Accurate knowledge state estimation can support learning resource recommendation, instructional intervention, and personalized learning path planning.

Early KT studies mainly used probabilistic models to describe knowledge acquisition and state transitions. Bayesian Knowledge Tracing (BKT) represents students' mastery of knowledge concepts through latent variables (Corbett & Anderson, 1994). Its individualized extension further considers differences among learners (Yudelsohn et al., 2013). With the development of deep learning, DKT uses recurrent neural networks to represent student knowledge as hidden states that evolve with interaction sequences (Piech et al., 2015). Subsequent studies extended

KT from the perspectives of interpretability, question semantics, and question difficulty. For example, Deep IRT incorporates item response theory into deep KT to improve model interpretability (Pardos & Heffernan, 2011; Yeung, 2019). EKT introduces question text to enhance exercise representations (Liu et al., 2021).

Recent KT studies have increasingly considered individual factors, such as student ability, question difficulty, emotional states, and learning behaviors. DIMKT dynamically estimates student knowledge states by modeling the effect of question difficulty (Shen et al., 2022). Dual State Personalized KT incorporates personalized states and emotional information to characterize student differences (Wang et al., 2025). These studies demonstrate the value of personalized information for knowledge state estimation. However, most existing methods model personalization through student ability, emotional factors, or general interaction patterns. They rarely consider individual differences in question understanding, knowledge concept association, and forgetting within a unified framework.

In real learning scenarios, students with similar response records may follow different cognitive processes when solving the same question, as shown in Fig. 1. First, prior knowledge, cognitive preferences,

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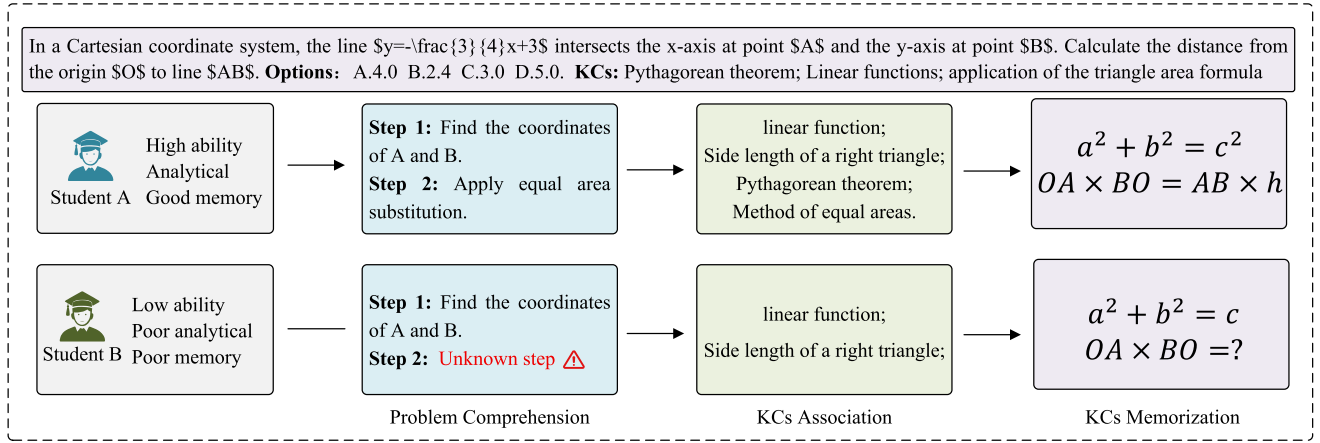


Fig. 1. Different students may interpret the same question in different ways.

and learning experience affect how students understand question semantics. Second, different interpretations may activate different knowledge concepts and produce different cognitive focuses. Third, learning ability, learning habits, and historical feedback influence forgetting rates and memory retention (Shen et al., 2024; Song et al., 2022). These factors interact and jointly affect the evolution of student knowledge states. Therefore, unified question representations and general interaction patterns cannot fully characterize personalized learning processes.

In addition to insufficient personalization, existing lightweight KT models have limited ability to mine deep semantic information. Many sequence-based models estimate student states mainly from question identifiers, knowledge concept identifiers, and response results (He et al., 2024; Piech et al., 2015). Text enhanced and question representation enhanced methods introduce additional exercise information. However, their representation capabilities still depend on predefined features or language encoders with limited capacity (Deng et al., 2025; Liu et al., 2021). Consequently, these methods may fail to capture complex semantic structures, latent conceptual relations, and cognitive information hidden in student interactions.

Large language models have strong semantic understanding and contextual modeling capabilities. They can extract rich latent information from question texts, knowledge concepts, and learning behaviors. However, their direct deployment requires substantial computational and storage resources (Fang et al., 2026; Shen et al., 2021). Knowledge distillation transfers informative representations from a large teacher model to a lightweight student model. It improves the representation capability of the student model while maintaining efficient inference (Yin et al., 2023a; Zhang et al., 2025). Therefore, integrating the semantic knowledge of large language models into lightweight KT models provides a practical balance between predictive performance and deployment efficiency. To address these limitations, this paper proposes Multi-level Individualized Perception and LLM-based Semantic-Enhanced Knowledge Tracing (MIPSEKT). The model contains two main components: multi-level personalized modeling and lightweight semantic enhancement. The personalized modeling component consists of three modules. The Personalized Question Understanding (PQU) module generates an individualized question representation by integrating student characteristics with question semantics. The Personalized Question-Knowledge Association Module (PQKA) module uses this representation to dynamically model the activation of different knowledge concepts. The Personalized Knowledge Forgetting (PKF) module incorporates interaction intervals, historical states, and student feedback to characterize individual differences in knowledge retention and forgetting.

For semantic enhancement, MIPSEKT employs a large language model as the teacher and an LSTM as the lightweight student. Hidden state distillation transfers deep semantic modeling capability from the

teacher to the student. Both models receive the same personalized interaction representations. Their hidden knowledge states are aligned at each time step during training. After training, only the lightweight student model is retained for prediction. This design improves semantic representation while maintaining low inference costs.

The contributions of our paper are:

- We propose a knowledge tracing framework that integrates multi-level personalized features. The framework models students' semantic understanding, the associations between questions and knowledge concepts, along with individual patterns of knowledge forgetting, thereby capturing individual differences in cognitive comprehension and knowledge mastery.
- To enhance semantic understanding, we introduce a knowledge distillation-based semantic knowledge enhancement mechanism. This mechanism leverages the powerful text semantic extraction and representation capability of large language models and transfers hidden-state semantic representations from the teacher LLM to the lightweight student model.
- Extensive experiments on two public datasets demonstrate the effectiveness of our model. The results show that, compared to baseline models, our approach achieves superior predictive performance, providing a solid foundation for personalized learning and education.

2. Related work

Accurate KT requires multi-level personalized modeling of student interaction information to characterize individual differences in understanding, association, and forgetting. It also requires enhanced semantic mining ability in the backbone model to fully exploit the deep information contained in question texts and KCs.

2.1. Knowledge tracing

KT aims to estimate learners' evolving knowledge states from their historical interactions. Early neural KT models mainly relied on recurrent neural networks to capture temporal learning patterns. For example, DKT represents students' knowledge states as hidden states that change over time (Piech et al., 2015). To improve interpretability, DKVMN introduces an external memory structure to explicitly model students' mastery of different KCs (Zhang et al., 2017). These methods laid the foundation for neural KT, but they mainly focus on sequential response patterns and provide limited modeling of fine-grained student differences.

With the development of attention mechanisms, many Transformer-based KT methods have been widely studied (Cheng et al., 2022;

Yin et al., 2023b). SAKT first applies self-attention to KT and captures the relevance between the current question and previous interactions (Pandey & Karypis, 2019). ExtraKT models short-term forgetting through a distance-based attention penalty, enabling effective extrapolation from short to long contexts (Li et al., 2024). AKT further introduces monotonic attention to reflect the temporal effect in learning sequences (Ghosh et al., 2020). Recent studies have also improved KT from structured attention, sparse modeling, and long-short term knowledge state separation. For example, DLSKT decouples long-term and short-term knowledge states and uses asymmetric knowledge distillation to promote their collaboration (Zhang & Zeng, 2026). These methods improve the ability to model learning sequences, but they still mainly describe general interaction dependencies rather than individualized learning processes.

Question texts and learning process information provide additional signals for knowledge state modeling. Text-aware KT methods use exercise descriptions to enhance question representations. For example, EERNN jointly models exercise records and exercise texts to improve student performance prediction (Su et al., 2018). BERT-based KT methods further extract richer semantic information from question texts and improve the quality of exercise representations (Li et al., 2023; Tong et al., 2020). In addition, response time, question difficulty, hints, feedback, and forgetting behavior can also reflect students' cognitive states. LFBKT models both learning and forgetting processes by incorporating question difficulty and auxiliary response features (Chen et al., 2022). KVFKT uses time interval-driven forgetting curves to describe knowledge decay (Guan et al., 2025). However, these methods usually model text semantics or forgetting factors separately, and they do not fully capture how different students understand the same question, activate related KCs, and forget knowledge in different ways.

Graph-based interaction modeling has also provided useful ideas for modeling complex relations. GKT methods employ graph neural networks to model concept relations and student proficiency (Nakagawa et al., 2021), while HCGKT further captures multi-level features through hierarchical graph learning and contrastive learning (Huang & Liu, 2025). In addition, DPNS reduces feedback conflict and false negatives in graph-based collaborative filtering through disentangled progressive negative sampling (Li et al., 2025). GIENN constructs a tag-aware interaction graph and uses attention mechanisms to model user-item interaction intentions (Dongjing et al., 2025). Although these studies are not designed for KT, they show that structured interaction modeling and intention-aware representation learning can help capture complex behavioral relations. SMDA-Net combines multiscale representation learning, efficient self-attention, and adaptive data augmentation for long-range time-series modeling (Zhang et al., 2024a). Inspired by these ideas, KT models still need more effective mechanisms to represent personalized interaction patterns and student-specific cognitive differences.

2.2. Semantic enhancement and knowledge distillation

Semantic information is important for understanding questions and student interactions in KT. Pretrained language models have therefore been introduced to improve semantic representation. For example, BiDKT uses BERT to model masked responses in interaction sequences and infer latent knowledge states (Tan et al., 2021). LBKT and BEKT combine BERT-based semantic representation with LSTM-based sequence modeling to handle long learning sequences (Tian et al., 2021; Xu et al., 2023). These studies show that semantic enhancement can improve KT performance. However, lightweight KT models still have limited ability to mine deep semantic information. Compared with BERT-based encoders, large language models have stronger semantic understanding and contextual reasoning ability, but directly deploying them in KT scenarios brings high computational and storage costs.

Knowledge distillation provides a feasible way to transfer useful knowledge from large models to lightweight models. Recent stud-

ies have explored distillation in different scenarios, including large model compression, temporal representation learning, and multimodal alignment. For example, GSDNet transfers knowledge from deep layers to shallow layers through gradual self-distillation and improves lightweight representation learning (Zhang et al., 2024c). KAID transfers multimodal interaction knowledge from a large CLIP teacher model to a lightweight student model through image feature matching and pixel-wise alignment (Zhang et al., 2024b). CTNet also shows that reconstructing crucial timestamps can enhance temporal representation learning in time series modeling (Zhang et al., 2025). These studies indicate that distillation can preserve useful semantic or structural knowledge while reducing deployment cost.

Some distillation methods further focus on how to transfer key knowledge more effectively. Difference-aware Distillation strengthens feature learning in highly discrepant regions through a dual-mask mechanism (Gou et al., 2024). MASCKD constructs sample correlations using multi-level attention maps to improve the student model's ability to absorb important knowledge (Gou et al., 2023). These works suggest that effective distillation should not simply imitate global features, but should guide the student model to learn informative representations from the teacher model.

Existing KT studies have improved knowledge state modeling through sequence modeling, attention mechanisms, semantic representation, and auxiliary information. Meanwhile, knowledge distillation provides a feasible way to transfer semantic knowledge from large models to lightweight models. However, how to jointly capture personalized learning differences and deep semantic information remains insufficiently explored. MIPSEKT uses an LLM as the teacher model and transfers hidden semantic representations to a lightweight LSTM student model. Different from existing semantic-enhanced KT methods, MIPSEKT does not only enhance question representations. It further combines LLM-based semantic distillation with multi-level personalized modeling. In this way, the student model can capture deep semantic information while also modeling individual differences in question understanding, knowledge activation, and forgetting.

3. Method

3.1. Problem definition

In KT, the DKT framework represents each student's learning process as a time-ordered sequence of questions and their corresponding responses. For a student, at time step t , the interaction information included the question, the related KCs, and the student's response. This was represented as a tuple (q_t, k_t, r_t) , where $q_t \in Q^+$ denoted the question, $k_t \in K^+$ denoted the KC, and $r_t \in \{0, 1\}$ denoted the student's response result, with 1 indicating a correct answer. Q^+ and K^+ represented the sets of questions and KCs, respectively. In addition, the framework incorporates used textual information and introduced additional interaction features $f_t \in F^+$, such as the number of hints, response time, and number of revisions. For example, the interaction record $(q_t, k_t, f_t, 1)$ indicates that, at time step t , the student answered question q_t correctly. The question is associated with KCs k_t , and f_t represents the feedback information generated during the interaction.

3.2. The model overview

This section presents an overview of the MIPSEKT model, as shown in Fig. 2. MIPSEKT comprises two main components that contain four functional modules. These modules characterize students' knowledge states from two aspects: personalized cognitive modeling and semantic mining enhancement. For ease of description and to simplify the mathematical notation, this paper uses the data processing flow of one learner as an example. First, the Personalized Question Understanding Module (PQU) models differences in students' semantic understanding of the

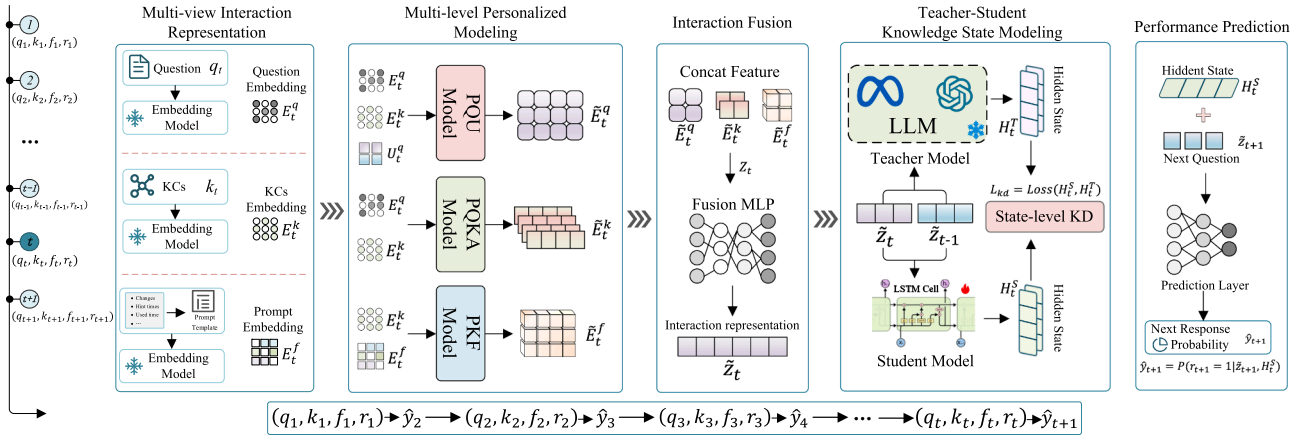


Fig. 2. The overall framework of the MIPSEKT model consists of two core components: the personalized module and the knowledge distillation-enhanced semantic module. The personalized module captures a student's individualized understanding of questions and the associated KCs, and models their forgetting characteristics based on feedback from interactions, thereby forming a comprehensive personalized student representation. The knowledge distillation-enhanced semantic module leverages a large language model to receive personalized student features, generate hidden states, and transfer this knowledge to enhance the hidden representations of the lightweight model. Here,  denotes trainable modules, and  denotes modules with frozen parameters.

same question. Second, the Personalized Question-Knowledge Association Module (PQKA) further captures individual differences in the association between questions and KCs. It obtains question-KC mappings that better match students' cognitive characteristics. Third, the Personalized Knowledge State Forgetting Module (PKF) models students' forgetting patterns over time by considering individual differences in the learning process. Finally, the Lightweight Semantic Mining Enhancement Module (SME) uses knowledge distillation to transfer the deep semantic mining ability of a large language model, which serves as the teacher model, to the lightweight student model. This enhances the student model's semantic representation and reasoning ability. Through the cooperation of these modules, MIPSEKT provides a more fine-grained characterization of students' knowledge states from question understanding, knowledge association, forgetting modeling, and semantic enhancement. This improves the prediction performance of KT.

3.3. Personalized question understanding module (PQU)

To characterize personalized cognitive differences in real question-solving scenarios, this paper designs the Personalized Question Understanding Module (PQU), as shown in Fig. 3. Existing KT models usually use a unified question embedding to represent question information. This setting assumes that different students have the same semantic understanding of the same question. However, students differ in cognitive preferences, knowledge backgrounds, and prior experiences. Therefore, their understanding processes and attention patterns for the same question may also differ. If such differences are ignored, the model may fail to capture student-level personalized semantic features. To address this issue, PQU explicitly models personalized understanding representations of question information for each student. This helps characterize the student's actual understanding state in the current interaction more accurately.

The question text at time step t is denoted as q_t . It is encoded by the text embedding module to obtain the question semantic representation $E_t^q \in \mathbb{R}^d$. Different students may have different semantic understandings of the same question because they differ in prior knowledge, cognitive preferences, and information extraction ability. To model such personalized differences, we introduce a student understanding ability representation $E_t^U \in \mathbb{R}^d$. It represents the individual characteristics of the student in question understanding and semantic information extraction. This representation is learnable during training and is continuously optimized as the student's knowledge state changes. The mathematical

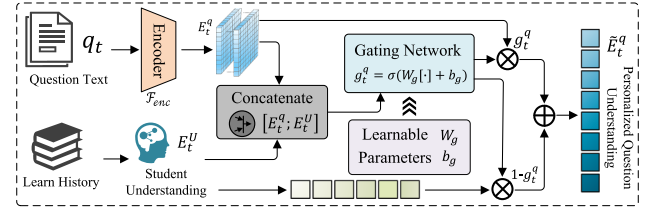


Fig. 3. Personalized question understanding module for a student.

formula is described as follows:

$$E_t^q = F_{enc}(q_t), \quad (1)$$

where $F_{enc}(\cdot)$ denotes the text embedding encoder, and $E_t^q \in \mathbb{R}^d$ denotes the semantic representation of question q_t at time step t .

$$g_t^q = \sigma(W_g[E_t^q; E_t^U] + b_g), \quad (2)$$

where $[\cdot; \cdot]$ denotes vector concatenation, W_g and b_g are learnable parameters, $\sigma(\cdot)$ denotes the sigmoid function, and g_t^q denotes the gating weight, which is used to adaptively adjust the contributions of question semantic information and student understanding ability information.

$$\tilde{E}_t^q = g_t^q \odot E_t^q + (1 - g_t^q) \odot E_t^U, \quad (3)$$

where $\odot$ denotes element-wise multiplication. $\tilde{E}_t^q$ denotes the personalized understanding representation of student $E_{(i,t)}^U$ for question q_t at time step t .

3.4. Personalized question-knowledge association module (PQKA)

In real learning scenarios, students may activate different KCs for the same question due to individual cognitive differences. In addition to knowledge mastery, cognitive style, thinking process, and attention allocation can also influence students' selective focus on relevant KCs. However, existing KT models usually adopt fixed question-KC mappings. They cannot fully reflect the personalized knowledge activation patterns formed when students understand questions. This limitation restricts the fine-grained modeling of students' cognitive states and learning processes (Fig. 4).

To address this issue, this paper proposes the Personalized Question-Knowledge Association module (PQKA). This module uses the student's question understanding representation E_t^q as the query. It dynamically computes its relevance to the vectors of all KCs E_t^k . The relevance scores

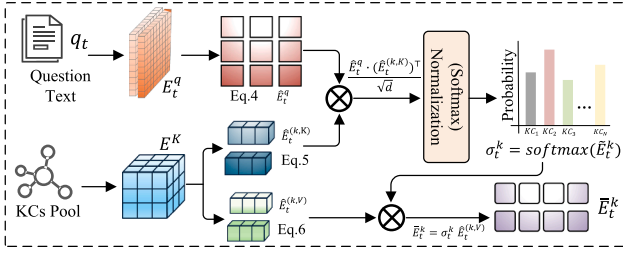


Fig. 4. Knowledge-concept association module following a student's question understanding .

measure the activation degree of different KCs after the student understands the current question. In this way, the module forms a personalized question–knowledge association representation.

$$\hat{E}_t^q = W_q E_t^q + b_q, \quad (4)$$

$$\hat{E}_t^{(k,K)} = W_k E_t^{(k,K)} + b_k, \quad (5)$$

$$\hat{E}_t^{(k,V)} = W_v E_t^{(k,V)} + b_v, \quad (6)$$

where W_q , W_k , and W_v are learnable parameter matrices, and b_q , b_k , and b_v are bias terms. $\hat{E}_t^q$ denotes the personalized question query vector of student at time step t , $\hat{E}_t^{(k,K)}$ and $\hat{E}_t^{(k,V)}$ denote the key and value representations of the KCs set after linear mapping, respectively.

Then, the matching scores between the personalized question understanding representation and all KCs are computed as follows:

$$\tilde{E}_t^k = \frac{\hat{E}_t^q \cdot (\hat{E}_t^{(k,K)})^\top}{\sqrt{d}}, \quad (7)$$

where $\tilde{E}_t^k$ denotes the relevance score between student and the i th KC after the student understands question q_t .

To obtain normalized personalized KCs activation weights, the softmax function is applied as follows:

$$\alpha_t^k = \text{softmax}(\tilde{E}_t^k), \quad (8)$$

where α_t^k denotes the personalized activation distribution of student over all KCs at time step t .

Finally, the KCs value representations are weighted and aggregated according to this activation distribution to obtain the personalized question–knowledge association representation:

$$\bar{E}_t^k = \alpha_t^k \hat{E}_t^{(k,V)}, \quad (9)$$

where $\bar{E}_t^k$ denotes the personalized question–knowledge association representation of student at time step t . This representation characterizes the association strength between the current question and different KCs after the student understands the question.

3.5. Personalized knowledge state forgetting module (PKF)

In KT tasks, modeling the decay of students' knowledge states over time or through interactions is a key factor for improving prediction accuracy. Traditional models such as DKT and AKT often neglect forgetting or model it in a simplistic manner, which fails to capture the underlying cognitive patterns.

According to Ebbinghaus' forgetting curve, memory strength decays exponentially over time. Cognitive diagnostic models emphasize that a student's mastery is a latent variable influenced by cognitive structure and learning history. Therefore, we introduce an explicit forgetting module that applies decay to the student's knowledge state after each interaction, aligning with natural forgetting processes while accounting for the reinforcing effects of repeated practice or review behavior.

Consider a student interacting with question q_t at time step t , with the latent knowledge mastery state represented by the vector $p_t \in \mathbb{R}^d$. We

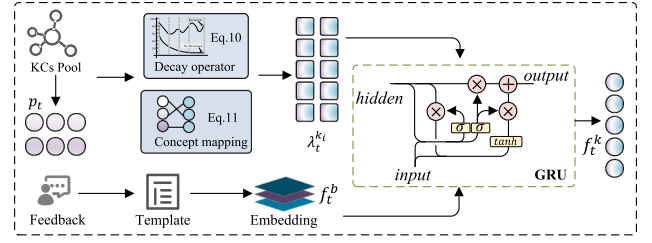


Fig. 5. The personalized forgetting representation is constructed through a dual-layer mechanism that integrates a forgetting curve and student feedback information.

define a memory retention factor to control the forgetting rate, which follows the formula below:

$$p_t^{\text{decay}} = p_t \odot \exp(-\gamma_t \Delta t), \quad (10)$$

where p_t^{decay} denotes the decayed knowledge state; $\gamma_t \in \mathbb{R}^d$ is the forgetting rate vector corresponding to each knowledge dimension; Δt represents the time interval, which is obtained from the statistics of students' historical sequences; $\odot$ represents element-wise multiplication; $\exp(-\gamma_t \Delta t) \in (0, 1]^d$ controls the degree of knowledge retention, following the exponential decay pattern proposed by Ebbinghaus.

The forgetting curve mapping corresponding to the knowledge concepts k_i can be formulated as:

$$\lambda_t^{k_i} = \text{mapping}(p_t^{\text{decay}}) W_{k_i}, \quad (11)$$

where W_{k_i} is a learnable parameter matrix used to linearly transform and regulate the forgetting state representation.

To further enhance personalized forgetting modeling, this paper introduces student feedback as an additional input to the model. During question solving, students generate various behavioral signals, such as perceived question difficulty, hint usage, and response time. These feedback signals reflect students' understanding of the question and related KCs from a behavioral perspective. They also provide useful evidence for modeling individual differences in forgetting. Inspired by prompt learning, this paper designs a feedback representation method based on a prompt template. This method integrates discrete feedback information into structured text and further maps it into embedding features that can be processed by the model. In this way, the model can interpret and utilize student feedback more effectively during training (Fig. 5).

An example of the designed prompt template is presented below.

template = <|start_prompt|>Context : This dataset contains records of students' feedback information.

- The question ID : {q_id}, and the text is {q_text};
- The question's Knowledge concepts : {kcs};
- The student changed their answer {changes} times while working on the question.
- The student spent {timestamp} seconds/minutes working on the question.
- The student used {hints} hints while working on the question.
- The difficulty level of this question is {difficulty}.

<|end_prompt|>

The structured feedback is first encoded by an embedding layer to obtain a continuous semantic representation, which serves as a unified and learnable feature for subsequent modeling and inference.

$$f_t^b = F_{\text{enc}}(F_t^{\text{template}}), \quad (12)$$

where F_{enc} denotes the embedding network for modeling feedback information, F_t^{template} represents the student feedback input, and f_t^b is the resulting semantic embedding of the feedback.

To integrate information features from both $\lambda_i^{k_i}$ and f_i^b perspectives, a mapping network is used to map them into a unified space. Finally, feature fusion is performed.

$$\tilde{\lambda}_i^{k_i} = \text{Mapping}_{MLP}(\lambda_i^{k_i}) \quad (13)$$

$$\tilde{f}_i^b = \text{Mapping}_{MLP}(f_i^b) \quad (14)$$

$$f_{input} = \tilde{\lambda}_i^{k_i} + \tilde{f}_i^b. \quad (15)$$

To enhance the personalization of forgetting curve modeling, the forgetting-related features and the student feedback representations are jointly fed into a student forgetting network. The personalized forgetting backbone is implemented using a GRU-based gated architecture to capture the temporal dynamics of student forgetting. The gated units dynamically regulate information flow based on the current input and historical states, selectively updating or retaining memory. This mechanism enables a more accurate characterization of students' personalized forgetting patterns across different learning stages.

$$f_i^k = \text{ForgeLayer}(f_{input}), \quad (16)$$

where f_i^k denotes the student forgetting state obtained through the dual modeling process.

After obtaining a student's personalized understanding of questions, the association between questions and KCs, and their forgetting state, we linearly integrate these three components to construct a comprehensive cognitive representation of the student. This representation supports accurate prediction of student responses in subsequent learning tasks.

$$h_t = \tilde{E}_i^q + \tilde{E}_i^k + f_i^k. \quad (17)$$

where $\tilde{E}_i^q$ denotes the personalized question understanding representation of the student at time step i . $\tilde{E}_i^k$ denotes the personalized question-knowledge association representation, and f_i^k denotes the personalized forgetting state representation. h_t denotes the comprehensive cognitive representation of the student in the current interaction.

This fusion method projects question understanding, KCs activation, and forgetting state into the same representation space. It provides the input for subsequent knowledge state modeling.

3.6. Lightweight semantic mining enhancement module (SME)

In natural language processing (NLP), large language models, such as LLaMA2-7B (Touvron et al., 2023) and BERT (Devlin et al., 2019), have strong semantic understanding capabilities and can handle complex language tasks. However, their high computational and deployment costs limit practical deployment. In contrast, lightweight models, such as LSTM, are more computationally efficient, but they have limited ability to capture semantic details. In KT tasks, these models often fail to fully model complex contexts and reasoning information.

To address the limited semantic modeling ability of lightweight models, this paper introduces a knowledge distillation mechanism. It transfers semantic knowledge from a large language model to a lightweight student model. This improves the semantic mining ability of the student model while maintaining computational efficiency. Unlike traditional distillation methods that mainly imitate the output probability distribution of the teacher model, this paper focuses on the deep semantic information contained in the hidden states of the teacher model. By distilling hidden states, the student model can obtain richer semantic representations. This improves its semantic modeling ability in KT tasks.

The student's interaction data are first processed through a personalization module to obtain a personalized representation. Then, a linear fusion strategy is applied to integrate the student's understanding of the question, the associated KCs, and the forgetting state of the knowledge, resulting in the final representation h_t .

KT has clear temporal characteristics. The current knowledge state of a student is related not only to the current interaction but also to the historical learning process. Therefore, before feeding h_t into the teacher

model, this paper adds positional encoding P_t to preserve temporal position information in the interaction sequence:

$$z_t = h_t + P_t, \quad (18)$$

where P_t denotes the positional encoding at time step t , and z_t denotes the teacher model input after incorporating positional information. By introducing positional encoding, the teacher model can distinguish interactions at different time steps. It can better model the temporal evolution of students' knowledge states.

The parameters of the teacher model are frozen, with student learning features embedded into the model. Then, z_t is fed into the teacher model for semantic mining and temporal modeling:

$$h_t^{tea} = \mathcal{T}_{tea}(z_t), \quad (19)$$

where $\mathcal{T}_{tea}(\cdot)$ denotes the teacher model, and h_t^{tea} denotes the hidden knowledge state of the student output by the teacher model at time step t . The teacher model uses its contextual semantic modeling ability to extract high-level semantic features from the current interaction and its temporal position. It therefore obtains a more expressive representation of the student's knowledge state.

Consistent with the teacher model, the student model also takes the comprehensive cognitive representation with positional information, z_t , as input. This setting ensures that the teacher and student models learn the student's knowledge state under the same interaction semantics. It also makes hidden state distillation more stable and interpretable. The hidden knowledge state of the student model is defined as follows:

$$h_t^{stu} = \mathcal{T}_{stu}(z_t), \quad (20)$$

where $\mathcal{T}_{stu}(\cdot)$ denotes the lightweight student model. In this paper, LSTM is used as the KT backbone of the student model. h_t^{stu} denotes the hidden knowledge state output by the student model at time step t . Compared with the teacher model, the student model has lower computational cost, but its semantic mining ability is limited.

To transfer deep semantic knowledge from the teacher model to the student model, this paper uses a hidden state distillation constraint. This constraint encourages the hidden knowledge state of the student model to approach that of the teacher model. Since the hidden states of the teacher and student models may have different dimensions, this paper first projects them into a shared representation space through linear mappings:

$$\bar{h}_t^{tea} = W_{tea} h_t^{tea} + b_{tea}, \quad (21)$$

$$\bar{h}_t^{stu} = W_{stu} h_t^{stu} + b_{stu}, \quad (22)$$

where W_{tea} , W_{stu} , b_{tea} , and b_{stu} are learnable parameters. $\bar{h}_t^{tea}$ and $\bar{h}_t^{stu}$ denote the projected teacher hidden state and student hidden state, respectively.

Then, hidden state distillation is performed by minimizing the mean squared error between the two projected representations:

$$\mathcal{L}_{kd} = \frac{1}{T} \sum_{t=1}^T \|\bar{h}_t^{stu} - \bar{h}_t^{tea}\|_2^2, \quad (23)$$

where $\mathcal{L}_{kd}$ denotes the hidden state distillation loss. Through this loss constraint, the student model can learn the implicit knowledge of the teacher model in semantic mining and contextual modeling. This enhances the knowledge state representation ability of the lightweight model.

3.7. Prediction layer

After hidden state distillation, the hidden state h_t^{stu} output by the student model is regarded as the student's knowledge state representation at time step t . This representation integrates personalized question understanding, KCs association, forgetting state, and semantic knowledge transferred from the teacher model.

To predict the student's response performance at the next time step, this paper jointly models the student hidden knowledge state and the representation of the next question:

$$h_{t+1} = [h_t^{stu}; \tilde{E}_{t+1}^q; \bar{E}_{t+1}^k], \quad (24)$$

where $\tilde{E}_{t+1}^q$ denotes the personalized question understanding representation of the next question, and $\bar{E}_{t+1}^k$ denotes the KCs representation corresponding to the next question. $[\cdot; \cdot; \cdot]$ denotes vector concatenation. h_{t+1} denotes the joint representation used to predict the next response.

Then, the prediction layer computes the probability that the student answers correctly at the next time step:

$$\hat{y}_{t+1} = \sigma(W_p h_{t+1} + b_p), \quad (25)$$

where W_p and b_p are the parameters of the prediction layer, and $\sigma(\cdot)$ denotes the sigmoid function. $\hat{y}_{t+1}$ denotes the predicted probability that the student correctly answers the question at time step $t + 1$.

The model is trained using the binary cross-entropy loss:

$$\mathcal{L}_{pred} = -\frac{1}{T} \sum_{t=1}^T [y_{t+1} \log(\hat{y}_{t+1}) + (1 - y_{t+1}) \log(1 - \hat{y}_{t+1})], \quad (26)$$

where $y_{t+1} \in \{0, 1\}$ denotes the ground-truth response at time step $t + 1$. The final training objective combines the prediction loss and the hidden state distillation loss:

$$\mathcal{L}_{total} = \mathcal{L}_{pred} + \beta \mathcal{L}_{kd}, \quad (27)$$

where $\mathcal{L}_{total}$ represents the total loss. β is the loss coefficient, which controls the balance between prediction and distillation. After training, only the student model is used for inference. This maintains computational efficiency while improving semantic modeling ability.

4. Experiment

4.1. Datasets

To validate the effectiveness of the proposed method, this paper employed two datasets containing rich textual features and feedback information: DBE-KT22 (Abdelrahman et al., 2022) and XES3G5M (Liu et al., 2023).

DBE-KT22¹ was constructed based on real student exercise practice activities in the Relational Databases course at the Australian National University (ANU). It includes undergraduate students from different disciplines, such as computer science, engineering, business, economics, and social sciences, and spans the period from 2018 to 2021.

XES3G5M² created by TAL Education Group, by contrast, records more than 5 million interactions of third-grade students in mathematics. The dataset is in Chinese and was collected from the interactions of 18,000 students with 8000 mathematics questions.

Both datasets contain students' feedback information. In addition to response results, the datasets also contain rich learning behavioral feedback information, including question difficulty ratings, answer confidence ratings, answer revision counts, and response time. These features effectively reflect students' learning states and answering confidence, providing rich support for personalized forgetting modeling and KT research (Table 1).

4.2. Baseline models

- DKT (Piech et al., 2015) uses RNNs to model temporal dependencies in students' learning, capturing the evolution of knowledge states.

Table 1

Dataset information about DBE-KT22 and XES3G5M. #Tokens denotes the number of tokens used for model training.

	DBE-KT22	XES3G5M
#Students	1361	18,066
#KCs	98	865
#Questions	212	7652
#Interactions	167,222	5,549,635
Language	English	Chinese
#Tokens	13M	271M

- DKVMN (Zhang et al., 2017) implements a dynamic key-value memory network, where static matrices store knowledge concepts and dynamic matrices update mastery levels, enhancing the modeling of concept relationships.
- Deep-IRT (Yeung, 2019) builds on DKVMN and incorporates item response theory (IRT) to account for question difficulty, enhancing interpretability.
- GKT (Nakagawa et al., 2021) represents knowledge as a graph and reformulates KT as a node-level time series classification using graph neural networks.
- SAKT (Pandey & Karypis, 2019) employs a self-attention mechanism to identify key KCs from past interactions.
- AKT (Ghosh et al., 2020) utilizes a monotonic attention mechanism to build context-aware representations of student interactions, capturing performance over appropriate time scales.
- EKT (Liu et al., 2021) extends EERNN by using a knowledge state matrix, which captures the impact of exercises on multiple concepts, while a memory network tracks concept mastery.
- MLBKT (Li et al., 2023) integrates multiple explicit features and their latent relationships using BERT to capture skill mastery, learning transfer, and question difficulty.
- LBKT (Xu et al., 2023) incorporates learners' behavioral features—speed, attempts, and hints—to quantify their effects on knowledge acquisition.
- ExtraKT (Li et al., 2024) introduces bias terms to simulate forgetting and adapts attention computation, enabling it to generalize from short- to long-window interactions.
- KVFKT (Guan et al., 2025) combines attention-based embeddings with forgetting-curve modeling to dynamically update knowledge states and improve prediction accuracy.

4.3. Implementation details

Previous research mainly relies on ID-based data. To ensure experimental fairness, we consistently use textual data and re-evaluate the baseline models. For the teacher model, we use LLaMA2-7B.³ The sequence length of student interactions is 200, and the text length is 128. Sequences shorter than this length are padded to meet the required length. The word embedding dimension is 4096, which is linearly projected to a 256-dimensional space, and the student model hidden size is 128. A dropout rate of 0.2 is applied to prevent overfitting. The batch size is 32, and the learning rate is 0.003. For all datasets, we randomly split 70% of the data for training, 10% for validation, and the remaining 20% for testing, ensuring that splits are performed at the individual learner level. We implement all models using the PyKT⁴ toolkit and optimize them with Adam for 200 epochs. The teacher model uses the default configuration of LLaMA2-7B, which includes 32 attention heads, 32 hidden layers, and a text embedding dimension of 4096, and is loaded in float16 precision to reduce GPU memory consumption.

¹ <https://dataverse.ada.edu.au/dataset.xhtml?persistentId=doi:10.26193/6DZWOH>.

² <https://github.com/ai4ed/XES3G5M>.

³ <https://huggingface.co/meta-llama/Llama-2-7b>.

⁴ <https://github.com/pykt-team/pykt-toolkit>.

Table 2

Performance comparison between baseline models and MIPSEKT. The performance metrics are AUC, ACC, and RMSE. The models are trained for up to 200 epochs with an early-stopping patience of 10 epochs.

Model	DBE-KT22			XES3G5M		
	AUC $\uparrow$	ACC $\uparrow$	RMSE $\downarrow$	AUC $\uparrow$	ACC $\uparrow$	RMSE $\downarrow$
DKT (Piech et al., 2015)	0.7819 $\pm$ 0.0013	0.7902 $\pm$ 0.0011	0.3841 $\pm$ 0.0011	0.7852 $\pm$ 0.0012	0.8173 $\pm$ 0.0013	0.3834 $\pm$ 0.0010
DKVMN (Zhang et al., 2017)	0.7831 $\pm$ 0.0012	0.7926 $\pm$ 0.0011	0.3822 $\pm$ 0.0012	0.7792 $\pm$ 0.0013	0.8155 $\pm$ 0.0010	0.3802 $\pm$ 0.0010
Deep-IRT (Yeung, 2019)	0.7814 $\pm$ 0.0012	0.7879 $\pm$ 0.0011	0.3837 $\pm$ 0.0012	0.7807 $\pm$ 0.0011	0.8175 $\pm$ 0.0010	0.3789 $\pm$ 0.0011
GKT (Nakagawa et al., 2021)	0.7307 $\pm$ 0.0009	0.7283 $\pm$ 0.0012	0.3806 $\pm$ 0.0012	0.7727 $\pm$ 0.0012	0.8135 $\pm$ 0.0011	0.3786 $\pm$ 0.0012
SAKT (Pandey & Karypis, 2019)	0.7782 $\pm$ 0.0014	0.7903 $\pm$ 0.0012	0.3790 $\pm$ 0.0013	0.7693 $\pm$ 0.0015	0.8124 $\pm$ 0.0013	0.3725 $\pm$ 0.0014
AKT (Ghosh et al., 2020)	0.7984 $\pm$ 0.0012	0.7953 $\pm$ 0.0015	0.3766 $\pm$ 0.0013	0.8207 $\pm$ 0.0011	0.8273 $\pm$ 0.0011	0.3614 $\pm$ 0.0012
EKT (Liu et al., 2021)	0.8100 $\pm$ 0.0012	0.8112 $\pm$ 0.0011	0.3694 $\pm$ 0.0012	0.7783 $\pm$ 0.0013	0.8145 $\pm$ 0.0010	0.3580 $\pm$ 0.0010
MLBKT (Li et al., 2023)	0.8199 $\pm$ 0.0014	0.8018 $\pm$ 0.0014	0.3580 $\pm$ 0.0011	0.8246 $\pm$ 0.0012	0.8163 $\pm$ 0.0013	0.3553 $\pm$ 0.0011
LBKT (Xu et al., 2023)	0.8218 $\pm$ 0.0014	0.8203 $\pm$ 0.0012	0.3564 $\pm$ 0.0012	0.8234 $\pm$ 0.0010	0.8165 $\pm$ 0.0013	0.3501 $\pm$ 0.0011
ExtraKT (Li et al., 2024)	0.8303 $\pm$ 0.0015	0.8290 $\pm$ 0.0013	0.3489 $\pm$ 0.0012	0.8314 $\pm$ 0.0013	0.8283 $\pm$ 0.0014	0.3470 $\pm$ 0.0013
KVFKT (Guan et al., 2025)	0.8338 $\pm$ 0.0012	0.8312 $\pm$ 0.0011	0.3450 $\pm$ 0.0011	0.8336 $\pm$ 0.0014	0.8300 $\pm$ 0.0011	0.3465 $\pm$ 0.0012
MIPSEKT	0.8547 $\pm$ 0.0008	0.8429 $\pm$ 0.0011	0.3386 $\pm$ 0.0010	0.8572 $\pm$ 0.0014	0.8583 $\pm$ 0.0012	0.3360 $\pm$ 0.0011

The learner-performance prediction problem is formulated as a binary classification task, where the learner's answer is 1 if correct and 0 otherwise. Therefore, we use the area under the curve (AUC), accuracy (ACC), and root mean squared error (RMSE) to evaluate the performance of each model, which are widely used evaluation metrics in KT tasks, with higher values indicating better performance (except for RMSE, where lower values indicate better performance). We use PyTorch to run all models on NVIDIA RTX A6000 GPU with 48GB memory.

5. Results and analysis

As shown in Table 2, MIPSEKT achieves the best results on DBE-KT22 and XES3G5M in terms of AUC, ACC, and RMSE. This indicates that the proposed method shows stable predictive performance across different datasets. On DBE-KT22, MIPSEKT obtains an AUC of 0.8547, an ACC of 0.8429, and an RMSE of 0.3386. Compared with the strongest baseline, KVFKT, MIPSEKT improves AUC and ACC by 0.0209 and 0.0117, respectively, and reduces RMSE by 0.0064. On XES3G5M, MIPSEKT achieves an AUC of 0.8572, an ACC of 0.8583, and an RMSE of 0.3360. Compared with KVFKT, it improves AUC and ACC by 0.0236 and 0.0283, respectively, and reduces RMSE by 0.0105. These results show that MIPSEKT improves the discriminative ability and prediction accuracy while reducing the prediction error.

The comparison with different types of baseline models further supports the effectiveness of MIPSEKT. Early sequence-based methods, such as DKT, DKVMN, and Deep-IRT, mainly model knowledge state changes based on students' historical response sequences. However, they have limited ability to model question semantics, individual student differences, and forgetting processes. Therefore, their overall performance is relatively weak on both datasets. In contrast, MIPSEKT introduces personalized question understanding, personalized question-knowledge association, and personalized forgetting modeling. It can capture cognitive differences in student interactions from multiple aspects and thus achieves better prediction performance.

Compared with attention-based and advanced sequence modeling methods, such as SAKT, AKT, and ExtraKT, MIPSEKT also shows clear advantages. These methods can capture long-range dependencies in interaction sequences. However, they usually focus more on global patterns within sequences. They provide insufficient modeling of individual differences in question understanding, KCs activation, and forgetting behavior. MIPSEKT further introduces multi-level personalized modeling on the basis of sequence modeling. This enables more fine-grained characterization of the dynamic changes in students' knowledge states.

Compared with text-aware or auxiliary information-enhanced methods, such as EKT, MLBKT, LBKT, and KVFKT, MIPSEKT has two main advantages. First, MIPSEKT not only uses question texts and KCs, but also models personalized understanding of the same question by different

students. It further captures the resulting differences in KCs activation. Second, MIPSEKT incorporates student feedback into forgetting modeling. This helps the model characterize individualized knowledge decay more accurately. In particular, MIPSEKT outperforms KVFKT on both datasets. This indicates that relying only on time intervals or forgetting curves is insufficient to fully describe student forgetting behavior. Combining feedback information with personalized cognitive modeling can further improve the accuracy of knowledge state estimation.

In addition, MIPSEKT introduces a semantic distillation mechanism based on a large language model. This mechanism transfers the deep semantic representation ability of the teacher model to the lightweight student model. It enhances the student model's ability to model latent semantic information in question texts, KCs, and learning interactions while maintaining computational efficiency. Therefore, MIPSEKT achieves a better balance among semantic understanding, personalized modeling, and knowledge state prediction.

The standard deviations in Table 2 are generally on the level of 10^{-3} . This suggests that MIPSEKT has good stability across different runs. Overall, MIPSEKT outperforms representative existing methods on both datasets and across all evaluation metrics. These results verify the effectiveness of multi-level personalized modeling and semantic distillation enhancement.

6. Ablation study

To verify the effectiveness of each component in the proposed method, we conducted ablation experiments on the key modules of MIPSEKT.

6.1. Ablation study on different modules

To verify the effectiveness of the key modules in MIPSEKT, we conduct ablation experiments by removing the Personalized Question Understanding module (w/o PQU), the Personalized Question-Knowledge Association module (w/o PQKA), the Personalized Knowledge State Forgetting Module (w/o PKF), and the Knowledge Distillation module (w/o KD). The experiments are conducted on DBE-KT22 and XES3G5M. The results are shown in Table 3. After removing any module, AUC and ACC decrease on both datasets, while RMSE increases. This indicates that each module contributes positively to model performance.

First, removing PQU leads to a clear performance drop. On DBE-KT22, AUC decreases from 0.8547 to 0.8369, ACC decreases from 0.8429 to 0.8316, and RMSE increases from 0.3386 to 0.3457. A similar trend is observed on XES3G5M. This shows that students have different understandings of the same question. If a unified question representation is used, the model cannot fully capture student-level semantic understanding differences. PQU improves fine-grained knowledge state representation by modeling personalized question understanding.

Table 3
Ablation study results by removing different modules on DBE-KT22 and XES3G5M.

Model	DBE-KT22			XES3G5M		
	AUC $\uparrow$	ACC $\uparrow$	RMSE $\downarrow$	AUC $\uparrow$	ACC $\uparrow$	RMSE $\downarrow$
w/o PQU	0.8369 $\pm$ 0.0011	0.8316 $\pm$ 0.0014	0.3457 $\pm$ 0.0011	0.8394 $\pm$ 0.0013	0.8334 $\pm$ 0.0015	0.3438 $\pm$ 0.0012
w/o PQKA	0.8365 $\pm$ 0.0012	0.8317 $\pm$ 0.0016	0.3455 $\pm$ 0.0015	0.8358 $\pm$ 0.0011	0.8312 $\pm$ 0.0014	0.3460 $\pm$ 0.0016
w/o PKF	0.8287 $\pm$ 0.0015	0.8281 $\pm$ 0.0011	0.3488 $\pm$ 0.0014	0.8295 $\pm$ 0.0016	0.8290 $\pm$ 0.0010	0.3484 $\pm$ 0.0011
w/o KD	0.8373 $\pm$ 0.0013	0.8329 $\pm$ 0.0012	0.3404 $\pm$ 0.0013	0.8383 $\pm$ 0.0014	0.8394 $\pm$ 0.0015	0.3393 $\pm$ 0.0013
MIPSEKT	0.8547 $\pm$ 0.0008	0.8429 $\pm$ 0.0011	0.3386 $\pm$ 0.0010	0.8572 $\pm$ 0.0014	0.8583 $\pm$ 0.0012	0.3360 $\pm$ 0.0011

Table 4
Plug-and-play ablation results for multi-level personalized modeling. PQU: personalized question understanding module; PQKA: personalized question-knowledge association module; PKF: personalized knowledge state forgetting module.

Model	DBE-KT22			XES3G5M		
	AUC $\uparrow$	ACC $\uparrow$	RMSE $\downarrow$	AUC $\uparrow$	ACC $\uparrow$	RMSE $\downarrow$
DKT	0.7819 $\pm$ 0.0013	0.7902 $\pm$ 0.0011	0.3841 $\pm$ 0.0011	0.7852 $\pm$ 0.0012	0.8173 $\pm$ 0.0013	0.3834 $\pm$ 0.0010
DKT + PQU	0.8118 $\pm$ 0.0014	0.8129 $\pm$ 0.0014	0.3613 $\pm$ 0.0013	0.8141 $\pm$ 0.0015	0.8201 $\pm$ 0.0015	0.3569 $\pm$ 0.0015
DKT + PQKA	0.7939 $\pm$ 0.0016	0.8055 $\pm$ 0.0014	0.3704 $\pm$ 0.0016	0.8098 $\pm$ 0.0014	0.8180 $\pm$ 0.0014	0.3582 $\pm$ 0.0012
DKT + PKF	0.8020 $\pm$ 0.0015	0.8058 $\pm$ 0.0013	0.3675 $\pm$ 0.0014	0.8022 $\pm$ 0.0013	0.8196 $\pm$ 0.0016	0.3591 $\pm$ 0.0013
AKT	0.7984 $\pm$ 0.0012	0.7953 $\pm$ 0.0015	0.3766 $\pm$ 0.0013	0.8207 $\pm$ 0.0011	0.8273 $\pm$ 0.0011	0.3614 $\pm$ 0.0012
AKT + PQU	0.8198 $\pm$ 0.0015	0.8238 $\pm$ 0.0017	0.3532 $\pm$ 0.0014	0.8325 $\pm$ 0.0015	0.8359 $\pm$ 0.0013	0.3547 $\pm$ 0.0014
AKT + PQKA	0.8083 $\pm$ 0.0017	0.8161 $\pm$ 0.0015	0.3580 $\pm$ 0.0015	0.8233 $\pm$ 0.0015	0.8198 $\pm$ 0.0016	0.3568 $\pm$ 0.0013
AKT + PKF	0.8140 $\pm$ 0.0015	0.8202 $\pm$ 0.0014	0.3565 $\pm$ 0.0012	0.8279 $\pm$ 0.0014	0.8217 $\pm$ 0.0017	0.3590 $\pm$ 0.0015

Second, removing PQKA also reduces performance on both datasets. On XES3G5M, AUC decreases from 0.8572 to 0.8358, ACC decreases from 0.8583 to 0.8312, and RMSE increases from 0.3360 to 0.3460. This indicates that fixed question-KC mappings cannot fully reflect personalized knowledge activation patterns formed after students understand questions. PQKA dynamically models the association between questions and KCs based on personalized question understanding. It therefore improves the characterization of KCs activation.

Third, removing PKF causes the largest performance drop. On DBE-KT22, AUC decreases to 0.8287, ACC decreases to 0.8281, and RMSE increases to 0.3488. On XES3G5M, AUC and ACC decrease to 0.8295 and 0.8290, respectively, while RMSE increases to 0.3484. This suggests that forgetting modeling is important for KT tasks. PKF incorporates student feedback to model personalized forgetting. It can better reflect changes in students' knowledge states over time and across interactions.

Finally, removing KD also degrades model performance. Compared with the full model, w/o KD obtains lower AUC and ACC on both datasets. This indicates that the knowledge distillation module enhances the semantic modeling ability of the lightweight student model. By aligning the hidden states of the teacher and student models, KD transfers deep semantic knowledge from the large language model to the student model. This improves its ability to represent latent semantic information in question texts, KCs, and learning interactions.

Overall, the complete MIPSEKT achieves the best results on all metrics. This indicates that multi-level personalized modeling and semantic distillation enhancement are complementary. PQU and PQKA model student differences from question understanding and KCs activation, respectively. PKF further models personalized forgetting, while KD enhances the deep semantic mining ability of the lightweight model. These results verify the effectiveness and necessity of each module.

6.2. Plug-and-play analysis of personalized modeling

Table 4 reports the results after adding PQU, PQKA, and PKF as plug-in modules to DKT and AKT. Overall, the three modules improve performance to different degrees on both backbone models and both datasets. This indicates that they demonstrate independent utility and transferability. The improvements are especially clear on DKT. After adding the personalized modules, AUC and ACC increase, while RMSE decreases. This suggests that these modules can alleviate the limitation of traditional sequence models in modeling individual differences.

Table 5
Performance comparison under different values of loss weight parameter β .

Weight β	DBE-KT22			XES3G5M		
	AUC	ACC	RMSE	AUC	ACC	RMSE
0.1	0.8493	0.8334	0.3435	0.8496	0.8491	0.3478
0.3	0.8547	0.8429	0.3386	0.8572	0.8583	0.3360
0.5	0.8482	0.8340	0.3487	0.8488	0.8454	0.3531
0.7	0.8379	0.8296	0.3517	0.8415	0.8382	0.3653

On the DKT backbone, PQU brings the most evident improvement. On DBE-KT22, DKT + PQU improves AUC from 0.7819 to 0.8118 and ACC from 0.7902 to 0.8129. It also reduces RMSE from 0.3841 to 0.3613. PQKA and PKF also improve the performance of DKT in a stable manner. This shows that question-KC association modeling and forgetting modeling can enhance the representation of students' knowledge states.

On the AKT backbone, adding personalized modules also improves performance, but the improvement is smaller than that on DKT. This may be because AKT has already captured part of the sequence dependencies through the attention mechanism. Therefore, the personalized modules mainly provide complementary information about student-level cognitive differences. Among these variants, AKT + PQU performs well on most metrics. This indicates that personalized question understanding can further enhance question semantic representation.

Overall, PQU, PQKA, and PKF can be transferred to different KT backbones as effective plug-in modules. They enhance the model's capabilities in question understanding, knowledge association, and forgetting modeling, respectively. These results further verify the effectiveness of multi-level personalized modeling.

6.3. Sensitivity analysis of parameter

To analyze the effect of the loss weight parameter β on model performance, this study further conducts a parameter sensitivity experiment. Specifically, β is varied over a predefined range, and the corresponding changes in prediction performance are observed. This analysis evaluates the influence of β on the overall learning process and final results. The results are shown in Table 5.

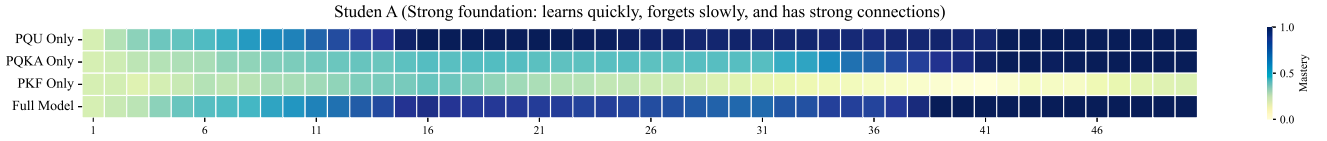


Fig. 6. Effects of individual personalized modules on student A's estimated mastery.

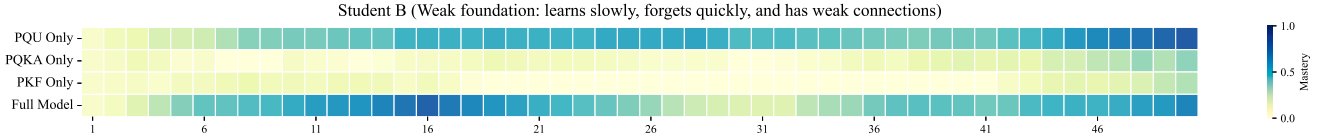


Fig. 7. Effects of individual personalized modules on student B's estimated mastery.

When $\beta = 0.3$, the model achieves the best results on both datasets. On DBE-KT22, the AUC, ACC, and RMSE are 0.8547, 0.8429, and 0.3386, respectively. On XES3G5M, the three metrics are 0.8572, 0.8583, and 0.3360, respectively. This indicates that a proper increase in β can strengthen the guidance provided by the distillation constraint during training. It can also improve the quality of knowledge state representations.

When β is small, such as $\beta = 0.1$, the model performs slightly worse than the best setting. This suggests that the auxiliary supervision signal is insufficient. The student model may not fully use the additional semantic information. When β further increases to 0.5 and 0.7, the model performance decreases. The RMSE also increases substantially. This indicates that an excessively large β may make the model focus too much on the auxiliary constraint. As a result, the optimization of the main prediction task is weakened. Overall, $\beta = 0.3$ shows stable performance on both datasets. It provides a good balance between the prediction loss and the auxiliary constraint. Therefore, $\beta = 0.3$ is used as the default setting in this study.

7. Case studies and visualizations

7.1. Knowledge state under personalized modeling

The case study is shown in Figs. 6 and 7. We select two students with different learning characteristics and analyze their knowledge state changes under the same question scenario. A darker color indicates a higher estimated mastery level. Student A has a stronger foundation, learns faster, and forgets more slowly, as shown in Fig. 6. Under the PQU Only and PQKA only-settings, the model captures effective understanding of question semantics and related KCs at an early stage. Under the PKF only-setting, the estimated mastery level decreases over time, but the decline is relatively slow. The full model further integrates personalized question understanding, question-KC association, and forgetting state information. As a result, the estimated knowledge mastery of Student A remains high in the middle and later stages.

In contrast, Student B has a weaker foundation, learns more slowly, and forgets more quickly, as shown in Fig. 7. Under the PQU only-setting, the improvement in mastery is limited. This indicates that Student B has insufficient semantic understanding of the same question. Under the PQKA only-setting, the activation of related KCs is weak. This suggests that the association between the question and KCs is less stable. Under the PKF only-setting, the mastery level remains low, indicating a stronger forgetting effect. Although the full model integrates multiple types of personalized information and improves the estimated knowledge state, the overall mastery level of Student B is still lower than that of Student A.

This case shows that students may exhibit different learning performance for the same question due to differences in semantic understanding, KCs association, and forgetting patterns. It further verifies the necessity of multi-level personalized modeling in MIPSEKT.

7.2. Semantic mining distillation visualization

As shown in Fig. 8, we visualize the effect of the knowledge distillation module on the semantic modeling ability of the lightweight student model. Fig. 8(a) shows the text semantic representation distributions of the teacher model, the vanilla LSTM, and the distilled LSTM. The teacher model forms clear clusters in the semantic space. This indicates that the large language model can extract stable deep semantic information from question texts and interaction contexts. In contrast, the representations of the vanilla LSTM are more dispersed and show a clear deviation from the teacher model. This suggests that the vanilla LSTM has limited ability to mine text semantics and latent conceptual relations. After knowledge distillation, the representations of the distilled LSTM move closer to the semantic clusters of the teacher model. This shows that distillation transfers the semantic structures learned by the teacher model to the lightweight student model.

Fig. 8(b) further shows the temporal hidden-state similarity between different student models and the teacher model. The similarity between the vanilla LSTM and the teacher model decreases rapidly as the time step increases. This indicates that the vanilla LSTM has difficulty maintaining semantic consistency with the teacher model during long-sequence modeling. In contrast, the distilled LSTM maintains a high similarity to the teacher model across the sequence. This suggests that the distillation constraint improves the alignment of contextual semantics and temporal knowledge states. The shaded area reflects the representation gain brought by knowledge distillation.

These results show that knowledge distillation improves not only the text semantic representation ability of the lightweight model but also its stability in sequential state modeling. By learning deep semantic representations from the teacher model, the student model can better capture latent information in question semantics, KCs relations, and learning behaviors. This provides a more reliable representation basis for subsequent knowledge state prediction.

7.3. Efficiency and computational cost comparison

To further evaluate the practical deployability of MIPSEKT, we compare different models in terms of parameter size, training time, inference speed, and inference VRAM usage. This experiment aims to verify whether the proposed distillation-based lightweight design can introduce the semantic capability of large language models while maintaining low computational cost and high inference efficiency.

Table 6 compares the computational cost and deployment efficiency of different models. The teacher model has strong semantic understanding and representation ability. However, it contains about 7.0B parameters and requires high training time and memory cost. Its inference speed is also relatively low. Therefore, directly deploying a large language model in online KT scenarios is impractical. In contrast, the BERT baseline reduces the parameter size and inference cost, but it still requires considerable training time and memory resources.

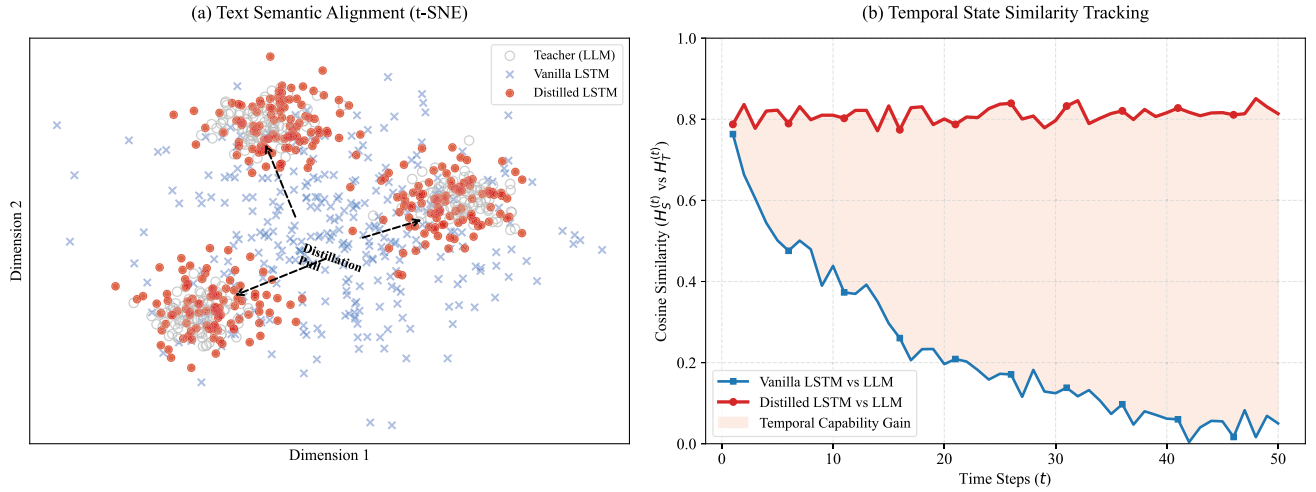


Fig. 8. Semantic representations of the teacher model, vanilla LSTM, and distilled LSTM during training.

Table 6
Efficiency and computational cost comparison of different models on same device.

Model	Params	Train Time / Epoch	Inference Speed	Inference VRAM
Teacher (LLM, e.g., LLaMA2-7B)	~7.0 B	~2.5 hrs	12 samples/s	~14.5 GB
Baseline (BERT)	110 M	~45 mins	185 samples/s	~1.2 GB
Vanilla Student (LSTM)	1.5 M	4 mins	4200 samples/s	~85 MB
Student (MIPSEKT)	1.8 M	7 mins	3600 samples/s	~85 MB

The lightweight LSTM student model shows clear advantages in efficiency. The vanilla student model contains only about 1.5M parameters. It requires about 4 min for each training epoch, achieves an inference speed of 4200 samples/s, and uses about 85 MB inference VRAM. After introducing personalized modeling and knowledge distillation, MIPSEKT maintains a similar parameter scale and inference VRAM usage. Its training time increases to about 7 min per epoch. This increase mainly comes from the additional computation of the distillation constraint and personalized modules. However, this cost only occurs during training. During inference, MIPSEKT uses only the lightweight student model. Therefore, it does not increase the actual deployment cost.

These results show that MIPSEKT introduces deep semantic information from the teacher model while maintaining high inference efficiency and low memory usage. This design avoids the high computational cost of directly deploying large language models. It also achieves a good balance between prediction performance and deployment feasibility. For online learning platforms that require real-time response and large-scale student interaction processing, MIPSEKT has strong practical value.

8. Conclusion

This paper proposes a KT framework that integrates multi-level personalized modeling and semantic enhancement. The framework uses student interaction data to characterize individual differences from three aspects: question understanding, question-KC association, and knowledge forgetting. Specifically, the Personalized Question Understanding module captures students' different semantic understandings of the same question. The personalized question-knowledge association module models personalized activation of related KCs. The Personalized Knowledge State Forgetting Module characterizes the dynamic changes in knowledge states during the learning process. To address the limited semantic representation ability of lightweight models, this paper further introduces knowledge distillation based on a large language model. It transfers deep semantic information from the teacher model to the student model. This improves the student model's representation capability and prediction stability while maintaining inference efficiency. Experi-

mental results show that the proposed framework can effectively model personalized learning processes and provide reliable support for knowledge state prediction in intelligent education scenarios.

CRedit authorship contribution statement

Zhenqiang Yu: Writing – original draft, Supervision, Methodology; **Luyao Huang:** Visualization, Data curation; **Xingbing Li:** Writing – review & editing, Formal analysis, Data curation; **Yuncheng Jiang:** Writing – review & editing, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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